

Lecture 2. Linear Regression Methods with Different Gradient Descent Algorithms

1. MSE Gradient

Gradient:

$$\nabla_{\mathbf{w}} Q(\mathbf{w}_k) = \frac{2}{\ell} X^T (X \mathbf{w}_k - \mathbf{y})$$

Learning rate:

$$\eta_k = \lambda \left(\frac{s_0}{s_0 + k} \right)^p$$

Linear regression:

$$\hat{\mathbf{y}} = X \mathbf{w}$$

Weight update:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \nabla_{\mathbf{w}} Q(\mathbf{w}_k)$$

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \frac{2}{\ell} X^T (X \mathbf{w}_k - \mathbf{y})$$

2. TimeDecayLR

Gradient:

$$\nabla_{\mathbf{w}} Q(\mathbf{w}_k) = \frac{2}{\ell} X^T (X \mathbf{w}_k - \mathbf{y})$$

Learning rate:

$$\eta_k = \lambda \left(\frac{s_0}{s_0 + k} \right)^p$$

where

$$s_0 = 1, \quad p = 0.5$$

Linear regression:

$$\hat{\mathbf{y}} = X \mathbf{w}$$

Weight update:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \nabla_{\mathbf{w}} Q(\mathbf{w}_k)$$

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \lambda \left(\frac{s_0}{s_0 + k} \right)^p \frac{2}{\ell} X^T (X \mathbf{w}_k - \mathbf{y})$$

3. StochasticGradientDescent

Gradient over a random batch B_k :

$$\nabla_{\mathbf{w}} Q_{B_k}(\mathbf{w}_k) = \frac{2}{|B_k|} X_{B_k}^T (X_{B_k} \mathbf{w}_k - \mathbf{y}_{B_k})$$

Learning rate:

$$\eta_k = \lambda \left(\frac{s_0}{s_0 + k} \right)^p$$

Linear regression:

$$\hat{\mathbf{y}}_{B_k} = X_{B_k} \mathbf{w}_k.$$

Linear regression weight update:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \frac{2}{|B_k|} X_{B_k}^T (X_{B_k} \mathbf{w}_k - \mathbf{y}_{B_k})$$

or, taking TimeDecayLR into account:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \lambda \left(\frac{s_0}{s_0 + k} \right)^p \frac{2}{|B_k|} X_{B_k}^T (X_{B_k} \mathbf{w}_k - \mathbf{y}_{B_k})$$

4. SAGDescent

Individual gradient for the i -th sample:

$$g_i(\mathbf{w}_k) = 2 (\mathbf{x}_i^T \mathbf{w}_k - y_i) \mathbf{x}_i$$

Average stored gradient:

$$\bar{\mathbf{g}}_k = \frac{1}{n} \sum_{i=1}^n \mathbf{g}_i$$

When the gradient of the selected sample j is recomputed, the average gradient is updated as follows:

$$\bar{\mathbf{g}}_{k+1} = \bar{\mathbf{g}}_k + \frac{1}{n} (\mathbf{g}_j^{new} - \mathbf{g}_j^{old})$$

Learning rate:

$$\eta_k = \lambda \left(\frac{s_0}{s_0 + k} \right)^p$$

Linear regression:

$$\hat{\mathbf{y}} = X\mathbf{w}.$$

Linear regression weight update:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \bar{\mathbf{g}}_{k+1}$$

or, taking TimeDecayLR into account:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \lambda \left(\frac{s_0}{s_0 + k} \right)^p \bar{\mathbf{g}}_{k+1}$$

5. MomentumDescent

MSE gradient:

$$\nabla_{\mathbf{w}} Q(\mathbf{w}_k) = \frac{2}{n} X^T (X\mathbf{w}_k - \mathbf{y})$$

Learning rate:

$$\eta_k = \lambda \left(\frac{s_0}{s_0 + k} \right)^p$$

Momentum accumulation:

$$\mathbf{h}_{k+1} = \alpha \mathbf{h}_k + \eta_k \nabla_{\mathbf{w}} Q(\mathbf{w}_k)$$

where

$$\mathbf{h}_0 = 0, \quad \alpha = 0.9.$$

Linear regression:

$$\hat{\mathbf{y}} = X\mathbf{w}.$$

For MSE:

$$\mathbf{h}_{k+1} = 0.9 \mathbf{h}_k + \eta_k \frac{2}{n} X^T (X \mathbf{w}_k - \mathbf{y})$$

Linear regression weight update:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \mathbf{h}_{k+1}$$

or, taking TimeDecayLR into account:

$$\mathbf{h}_{k+1} = 0.9 \mathbf{h}_k + \lambda \left(\frac{s_0}{s_0 + k} \right)^p \frac{2}{n} X^T (X \mathbf{w}_k - \mathbf{y})$$

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \mathbf{h}_{k+1}$$

6. Adam

Initial moment values:

$$\mathbf{m}_0 = \mathbf{0}, \quad \mathbf{v}_0 = \mathbf{0}$$

MSE gradient:

$$\mathbf{g}_k = \nabla_{\mathbf{w}} Q(\mathbf{w}_k) = \frac{2}{n} X^T (X \mathbf{w}_k - \mathbf{y})$$

Learning rate:

$$\eta_k = \lambda \left(\frac{s_0}{s_0 + k} \right)^p$$

First moment:

$$\mathbf{m}_{k+1} = \beta_1 \mathbf{m}_k + (1 - \beta_1) \mathbf{g}_k$$

Second moment:

$$\mathbf{v}_{k+1} = \beta_2 \mathbf{v}_k + (1 - \beta_2) \mathbf{g}_k^2$$

First moment correction:

$$\hat{\mathbf{m}}_{k+1} = \frac{\mathbf{m}_{k+1}}{1 - \beta_1^{k+1}}$$

Second moment correction:

$$\hat{\mathbf{v}}_{k+1} = \frac{\mathbf{v}_{k+1}}{1 - \beta_2^{k+1}}$$

Method parameters:

$$\beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \varepsilon = 10^{-8}$$

Linear regression:

$$\hat{\mathbf{y}} = X\mathbf{w}.$$

Weight update:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \frac{\hat{\mathbf{m}}_{k+1}}{\sqrt{\hat{\mathbf{v}}_{k+1}} + \varepsilon}$$

Taking TimeDecayLR into account:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \lambda \left(\frac{s_0}{s_0 + k} \right)^p \frac{\hat{\mathbf{m}}_{k+1}}{\sqrt{\hat{\mathbf{v}}_{k+1}} + \varepsilon}$$

Algorithm for Comparing Gradient Descent Methods and Selecting the Optimal Step Size

The following methods should be compared:

VanillaGradientDescent, StochasticGradientDescent, SAGDescent, MomentumDescent, A

For each method, all parameters except the initial step size λ should be kept at their default values.

1. Selecting the Optimal Step Size

For each method independently, we consider the values

$$\lambda_1, \lambda_2, \dots, \lambda_m$$

on a logarithmic grid, for example,

$$\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}.$$

For each value λ_j :

1. train the model on the training set;

2. compute the error on the validation set:

$$Q_{\text{val}}(\lambda_j) = \frac{1}{\ell_{\text{val}}} \|X_{\text{val}}w(\lambda_j) - y_{\text{val}}\|^2;$$

3. save the obtained error.

The optimal step size for a given method is defined as

$$\lambda^* = \underset{\lambda}{\operatorname{argmin}} Q_{\text{val}}(\lambda)$$

That is, the best value of λ is the one that minimizes the error on the validation set.

The test set must not be used when selecting λ^* .

2. Table for Selecting the Optimal λ

For each method, we construct a table of the following form:

Таблица 1: Comparison of Gradient Descent Methods

Method	λ^*	Loss train	Loss test	R^2_{train}	R^2_{test}	Iterations
Vanilla GD						
SGD						
SAG						
Momentum						
Adam						

The optimal step size is selected based on the column

$$\text{Validation error}$$

as the value of λ corresponding to the minimum error.

3. Plot for Step Size Analysis

For each method, we can plot

$$Q_{\text{val}}(\lambda)$$

where the x -axis represents

$$\lambda$$

on a logarithmic scale, and the y -axis represents

$$Q_{\text{val}}.$$

The optimal step size corresponds to the minimum of this plot:

$$\lambda^* = \arg \min_{\lambda} Q_{\text{val}}(\lambda)$$

This plot provides a clear visualization of the results obtained from the logarithmic grid search.

4. Final Training of Each Method

After the optimal value has been found for each method,

$$\lambda_{\text{Vanilla}}^*, \quad \lambda_{\text{SGD}}^*, \quad \lambda_{\text{SAG}}^*, \quad \lambda_{\text{Momentum}}^*, \quad \lambda_{\text{Adam}}^*,$$

the following values should be saved for each method:

$$Q_{\text{train}}, \quad Q_{\text{test}}, \quad R_{\text{train}}^2, \quad R_{\text{test}}^2, \quad N_{\text{iter}}.$$

5. Final Comparison Table of Gradient Descent Methods

Construct a single table:

Таблица 2: Final Comparison of Gradient Descent Methods

Method	Best λ^*	Loss train	Loss test	R_{train}^2	R_{test}^2	Number of iterations
VanillaGradientDescent						
StochasticGradientDescent						
SAGDescent						
MomentumDescent						
Adam						

6. Convergence Comparison Plot

Using the optimal value of λ^* already found for each method, construct a single plot showing

$$Q_{\text{train}}(w_k) \text{ versus the iteration number } k$$

for all five methods.

On the x -axis:

$$k = \text{iteration number.}$$

On the y -axis:

$$Q_{\text{train}}(w_k) = \frac{1}{\ell_{\text{train}}} \|X_{\text{train}} w_k - y_{\text{train}}\|^2.$$

The plot should contain five curves:

$$\text{Vanilla GD, \quad SGD, \quad SAG, \quad Momentum, \quad Adam.}$$

7. How to Draw the Final Conclusion

The best value of λ within each method is determined solely as

$$\lambda^* = \arg \min_{\lambda} Q_{\text{val}}(\lambda).$$

After selecting the optimal value λ^* , the methods are compared with each other.

Better performance on new data corresponds to a lower value of Q_{test} and a higher value of R^2_{test} .

The convergence speed is determined by the number of iterations N_{iter} and by the plot of the training error versus the iteration number, $Q_{\text{train}}(w_k)$.

Thus, the final comparison should take into account both model performance and convergence speed.